Terence – Ramanujan Game

Problem

Terence and Ramanujan take turns picking numbers from a bowl containing the first 25 Fibonacci numbers:

$$F_1 = 1, \quad F_2 = 1, \quad F_3 = 2, \quad \ldots, \quad F_{25} = 75025.$$

For a chosen number k, define

$$W(k) = k^{10} + k^5 + 1.$$

Let $P(k)$ denote the product of all proper divisors of $W(k)$.
A player wins a round if their chosen number k yields the smallest possible value of $P(k)$ among all available choices.
Find the probability that Terence wins on a randomly chosen pick.